

First Normal Form (1NF)

Introduction

First Normal Form (1NF) is the *most basic* level of database normalization. It ensures that the data in a table is stored in a *simple, organized, and unambiguous* manner.

Definition

A relation (table) is in **First Normal Form** if:

1. All attributes (columns) contain **atomic** values (no multiple or composite values in a single cell).
2. There are no **repeating groups** or arrays.
3. Each record is unique (there is a primary key).

Why 1NF is Important

- Prevents storing multiple values in a single cell.
- Makes data retrieval simpler and more efficient.
- Avoids anomalies during **INSERT**, **UPDATE**, and **DELETE** operations.

Example: Violation of 1NF

Consider the following table storing student course information:

Student_ID	Name	Courses
101	Alice	Math, Physics
102	Bob	Chemistry, Biology, Physics
103	Charlie	Math

Problems:

- The **Courses** column contains *multiple values* in a single cell.
- Difficult to query students enrolled in a specific course.
- Causes redundancy and inconsistency.

Transforming to 1NF

We split multi-valued attributes into separate rows.

Student_ID	Name	Course
101	Alice	Math
101	Alice	Physics
102	Bob	Chemistry
102	Bob	Biology
102	Bob	Physics
103	Charlie	Math

Key Points to Remember

- **Atomicity:** Each column must store indivisible values.
- **Uniqueness:** Each record must be identifiable by a key.
- **No Repetition:** Avoid repeating groups within a table.

Real-World Analogy

Think of 1NF as a **neatly organized bookshelf**:

- Each book (data item) is in its own slot (atomic value).
- No piles of mixed books in a single slot (no multi-valued fields).
- Every slot is labeled (unique key) so you can easily find a book.

Summary Table

Rule	Description
Atomic Values	Each cell should hold only one value.
No Repeating Groups	No lists or sets inside a column.
Unique Rows	Every row must be distinguishable using a primary key.

Tip

Always check for *multi-valued attributes* when designing your schema. Breaking them down into separate rows ensures compliance with 1NF and simplifies queries.